SAT Hard Question Set 15

Jack Zeng

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Phase 2 · Hard Question Practice

Overview

Hard Question Set 15

9 SAT-style challenge problems

Question 1

Which of the following ensures that the quadratic equation

$$ax^2 - 4x + c = 0$$

has at least one real root?

A. $a > 0$

B. $a = 0$

C. $c > 0$

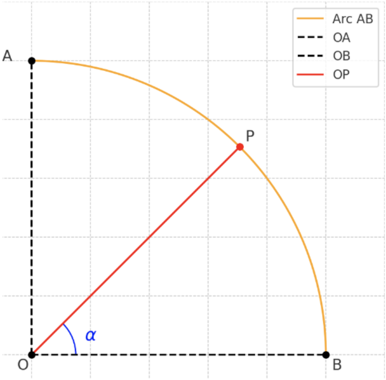
D. $c = 0$

Answer 1

Correct Answer: D

Question 2

As shown in the figure, O is the center of a circle with radius 1, and the circle intersects the coordinate axes at points A and B. Point P lies on arc $\overset{\frown}{AB}$ (not overlapping with A or B). Connect OP, and let $\angle POB = \alpha$. What are the coordinates of point P?



A. $(\sin\alpha,\ \sin\alpha)$

Answer 2

Correct Answer: C

Question 3

The following table shows the age distribution of a school choir's members:

Age (years)	13	14	15	16
Frequency	5	15	x	$10 - x$

For different values of x, which pair of statistical measures regarding age will not change?

A. Mean, Median

B. Mode, Median

C. Mean, Standard Deviation

D. Median, Standard Deviation

Answer 3

Correct Answer: B

Question 4

Given that the line $y = kx + k + 1$ passes through the points $(m,\ n+3)$ and $(m+1,\ 2n-1)$, and that $0 < k < 2$, what is a possible value of n?

A. 3

B. 4

C. 5

D. 6

Answer 4

Correct Answer: C

Question 5

The graph of the quadratic function $y = |a|x^2 + bx + c$ passes through the following five points:

$$A(m,\ n),\quad B(0,\ y_1),\quad C(3-m,\ n),\quad D(\sqrt{2},\ y_2),\quad E(2,\ y_3).$$

What is the correct order of y_1, y_2, y_3?

A. $y_1 < y_2 < y_3$

B. $y_1 < y_3 < y_2$

C. $y_3 < y_2 < y_1$

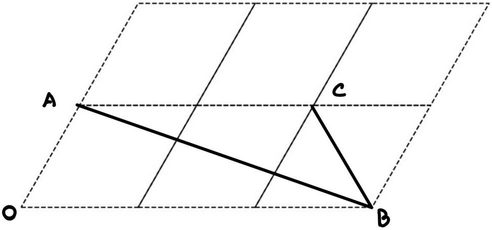
D. $y_2 < y_3 < y_1$

Answer 5

Correct Answer: B

Question 6

As shown in the figure, 6 identical rhombuses form a grid. The vertices of the rhombuses are called grid points. It is known that one angle of the rhombus, $\angle O$, is 60°, and points A, B, and C are located on grid points. Find the value of $\tan \angle ABC$.



Answer 6

Final Answer: $\boxed{\sqrt{3}/2}$

Question 7

Given the quadratic function

$$y = x^2 - 2ax + a \quad (a \neq 0)$$

passes through two points $A\left(\frac{a}{2},\ y_1\right)$ and $B(3a,\ y_2)$, which of the following statements is correct?

A. There exists a real number a such that $y_1 > a$.

B. For all real numbers a, $y_1 > a$ is always true.

C. There exists a real number a such that $y_2 < 0$.

D. For all real numbers a, $y_2 < 0$ is always true.

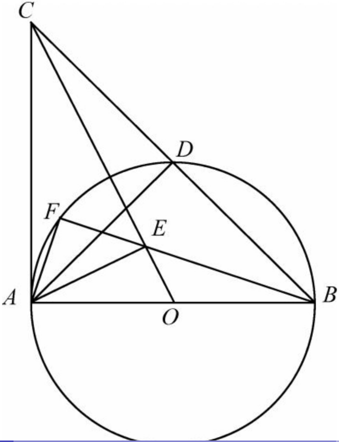
Answer 7

Correct Answer: C

Question 8

As shown in the figure, in triangle $\triangle ABC$, $\angle BAC = 90^\circ$, $AB = AC$. Circle O has diameter AB, and intersects BC at point D. Line $AE \perp OC$, with foot of the perpendicular at point E. The extension of line BE intersects AD at point F.

Find the value of $\dfrac{OE}{AE}$.

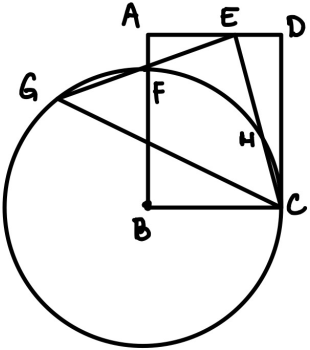


Answer 8

Final Answer: $\boxed{\frac{1}{2}}$

Question 9

In rectangle $ABCD$, point B is the center of a circle $\odot B$ with radius BC. The circle intersects AB at point F. Point E lies on AD, and segment CE is rotated about E to form EG such that point G lies on $\odot B$, where $\angle GEC = 90^\circ$. Point F is the midpoint of EG. Given that $AF = 1$ and $AE = 3$, find the length of CD.



Answer 9

Final Answer: $\boxed{6}$

Thank You!

We appreciate your time and focus.

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